Answers

Physics

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
1	(3)	Circular Motion	14	(3)	Wave Optics
2	(4)	Conversion of Galvanometer into Voltmeter	15	(1)	Optical instruments
3	(3)	Impulse	16	(3)	Atoms
4	(3)	Rotational Motion	17	(2)	Spectral Series
5	(3)	Gravitation	18	(4)	Matter Wave
6	(3)	Simple Harmonic Motion	19	(4)	Photoelectric Effect
7	(2)	Bulk Modulus of Elasticity	20	(4)	Semiconductor Devices
8	(3)	Capillary rise	21	[30]	Refraction through Prism
9	(1)	Inductance and AC Circuit	22	[375]	Adiabatic Process
10	(2)	Buoyancy and Archimedes' principle	23	[5.00]	Mass-Pulley System
11	(2)	Relation between Electric Field and Potential	24	[36.00]	Power in Electrical Circuits
12	(2)	Speed of Sound	25	[7.07]	Magnetic Effects of Current
13	(2)	Electromagnetic Waves			

Chemistry

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
26	(3)	Mix reaction	39	(3)	Carbonyl Compound
27	(1)	Chemical Bonding	40	(2)	Carboxylic Acid
28	(1)	Ionic Equilibrium	41	(2)	Nitrogen Compound
29	(1)	Periodic Table	42	(2)	Biomolecules
30	(4)	Chemical Equilibrium	43	(4)	Structure of atom
31	(2)	Chemical Energetics	44	(1)	Halogen Derivatives
32	(2)	Redox Reaction	45	(3)	Alcohol, Ether and Phenol
33	(2)	Chemical Bonding	46	[10.20]	Chemical Kinetics
34	(4)	d and f block	47	[6.0]	Dipole Moment
35	(2)	IUPAC	48	[266]	Electrochemistry
36	(1)	Coordination Compound	49	[112.70]	Solution
37	(4)	Coordination Compound	50	[1.12]	Aromatic Compound
38	(4)	Chemical Reactions of Ethers			

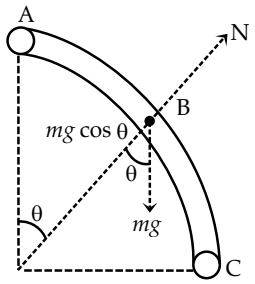
Mathematics

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
51	(2)	Binomial theorem	64	(3)	Binomial Theorem
52	(1)	Definite Integration	65	(4)	Integration
53	(3)	Determinants	66	(2)	Sets
54	(1)	Differential Equations	67	(1)	Parabola
55	(1)	Sraight Lines	68	(3)	Function
56	(3)	Vectors	69	(3)	Hyperbola
57	(1)	Matrices	70	(3)	Probability
58	(2)	Straight Lines	71	[80.00]	Trigonometric Identities
59	(2)	Probability	72	[8.00]	Inverse Trigonometric Functions
60	(3)	Complex Numbers	73	[8.00]	Limits
61	(4)	Straight Lines	74	[11.00]	Complex Number
62	(2)	Functions	75	[2.00]	Differentiation
63	(2)	Sequence and Series			

ANSWERS WITH EXPLANATION

Physics

1. Option (3) is correct



For circular motion of ball in the tube,

$$mg\cos\theta - N = \frac{mv^2}{r}$$

As Ball B moves down, the value of θ increases, cos θ decreases.

$$N = mg\cos\theta - \frac{mv^2}{r}$$

So, N continuously decreases and it becomes 0 when $\cos\theta = \frac{v^2}{rg}$. Thereafter, it becomes negative because the ball leaves contact with surface A and comes in contact with outer part C.

[for e.g., at lowest point we can say,

$$N = mg\cos 90° - \frac{mv^2}{r}$$

$$= -\frac{mv^2}{r} \rightarrow \text{contact with C]}$$

2. Option (4) is correct.

According to the question, $V = I_g(R + R_g)$.

Full scale deflection current of galvanometer:

$$I_g = 25 \times 4 \times 10^{-4}$$
$$= 100 \times 10^{-4}\ A$$
$$= 10^{-2}\ A$$

As $$R = \frac{V}{I_g} - R_g$$

$$= 25 \times 10^2 - 50$$
$$= 2450\ \Omega \text{ in series}$$

3. Option (3) is correct.

Initial momentum of the ball perpendicular to the wall,

$$P_i = m(-v\cos 60°)$$
$$P_i = \frac{-mv}{2}$$

Similarly, Final momentum of the ball perpendicular to the wall,

$$P_f = m(v\cos 60°)$$
$$P_f = \frac{mv}{2}$$

Now, Impulse = Change in momentum

$$I = P_f - P_i$$
$$I = \frac{mv}{2} - \left(\frac{-mv}{2}\right)$$
$$I = \frac{mv}{2} + \frac{mv}{2}$$
$$I = mv$$

4. Option (3) is correct.

Applying conservation of angular momentum about P

$$\Rightarrow \quad \frac{ML^2}{12}\omega = \frac{ML^2}{3}\omega_1$$

$$\Rightarrow \quad \omega_1 = \frac{\omega}{4}$$

$$\therefore \quad v_B = \frac{\omega L}{4}$$

5. Option (3) is correct.

$$g = \frac{GM}{R^2}$$

(Given R → Radius of Earth,
ρ → Mean density of Earth)

$$g = \frac{G\left(\frac{4}{3}\pi R^3\rho\right)}{R^2};$$

$$g = \frac{4}{3}\pi R\,\rho G$$

$$\rho = \frac{3}{4}\cdot\frac{g}{\pi RG}$$

6. Option (3) is correct.

Given $x = 7\cos 0.5\,\pi t$

General form of equation

$$x = A\cos\omega t$$

Given equation compare with standard form

$$\omega = \frac{2\pi}{T} = 0.5\,\pi;\ T = 4\ s$$

$$T' = \frac{T}{4} = \frac{4}{4} = 1\ s$$

7. Option (2) is correct.

$$\Delta P = h\rho g$$
$$= 200 \times 10^3 \times 10$$
$$= 2 \times 10^6\ N/m^2$$

$$K = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{2\times10^6}{\frac{1}{100}}$$

$$= 2\times 10^9 \text{ N/m}^2$$

8. Option (3) is correct.

$$h = \frac{2T\cos\theta}{r\rho g}$$

$$r_1 h_1 = r_2 h_2$$

and $$A = 2\pi r^2$$

$$\Rightarrow \quad r \propto \sqrt{A}$$

$$\therefore \quad \sqrt{A_1}\ h_1 = \sqrt{A_2}\ h_2$$

$$\Rightarrow \quad \sqrt{A} \times 4 = \sqrt{\frac{A}{4}} \times h_2$$

$$\Rightarrow \quad h_2 = 8 \text{ cm}$$

9. Option (1) is correct.

The insertion of the iron rod increases the inductance of the circuit, which in turn reduces the current flow and subsequently dims the bulb. Thus, Statement II provides a valid explanation for the observed effect described in Statement I.

10. Option (2) is correct.

Height of cone = 25 cm

Base diameter = 25 cm

Volume of the cone $= \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times 3.14 \times (12.5)^2 \times 25 \text{ cm}^3$$

$$= 4088 \text{ cm}^3$$

The volume of already immersed part of the cone

$$V' = \frac{1}{3} \times 3.14 \times (10)^2 \times 20$$

$$= 2093 \text{ cm}^3$$

Water to be displaced for complete immersion

$$= 4088 - 2093$$

$$= 1995 \text{ g}$$

$$\cong 2 \text{ kg}$$

Additional mass to be put

$$= 2000 \text{ g} = 2 \text{ kg}$$

11. Option (2) is correct.

Here, $\vec{E} = 5\hat{i} - 3\hat{j}$ kV/m

$$V_B - V_A = -\int_{r_A}^{r_B} \vec{E}.dr$$

$$= -\int_{(4,0,3)}^{(10,3,0)} \left(5\hat{i} - 3\hat{j}\right).\left(dx\hat{i} + dy\hat{j} + dz\hat{k}\right)$$

$$= -\int_4^{10} 5dx - \int_0^3 (-3)dy + 0 = -5\left[x\right]_4^{10} + 3\left[7\right]_0^3$$

$$= -5(10-4) + 3(3-0) = -30 + 9 = -21 \text{ kV}$$

12. Option (2) is correct.

As $$v = \sqrt{\frac{\gamma RT}{M}}$$

$$\therefore \quad \frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}}$$

13. Option (2) is correct.

$$E_0 = B_0 c$$

$$\Rightarrow \quad B_0 = \frac{E_0}{c}$$

$$\therefore \quad B = \frac{E_0}{c}\cos(\omega t - kz)\hat{j}$$

14. Option (3) is correct.

Here direction of light is given by normal vector

$$\hat{n} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$\therefore$ angle made by the $\hat{n}$ with y-axis is given by

$$\cos\beta = \frac{2}{\sqrt{1^2+2^2+3^2}} = \frac{2}{\sqrt{14}}$$

$$\beta = \cos^{-1}\frac{2}{\sqrt{14}}$$

15. Option (1) is correct.

$$A_I = m^2 A_0$$

$$\therefore \quad A_I = (4)^2 \times 100$$

$$\therefore \quad A_I = 1600 \text{ cm}^2$$

16. Option (3) is correct.

$$N = \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$$

17. Option (2) is correct.

The short limit of the Balmer series is given by

$$\bar{\nu} = \frac{1}{\lambda}$$

$$= R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4}$$

$$\therefore \quad R = \frac{4}{\lambda} = \frac{4}{3646} \times 10^{10} \text{ m}^{-1}$$

Further, the wavelengths of the Kα series are given by the relation:

$$\bar{\nu} = \frac{1}{\lambda}$$

$$= R(Z-1)^2\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$$

The maximum wave number correspnods to $n = \infty$ and therefore, we must have

$$\bar{\nu} = \frac{1}{\lambda} = R\,(Z-1)^2$$

or $$(Z-1)^2 = \frac{1}{R\lambda}$$

$$= \frac{3646\times10^{-10}}{4\times1\times10^{-10}}$$

$$= 911.5$$

$\therefore$ $$(Z-1) = \sqrt{911.5}$$

$$\cong 30.2$$

or $$Z = 31.2 \cong 31$$

Thus, the atomic number of the element concerned is 31. The element having atomic number Z = 31 is Gallium.

18. Option (4) is correct.

Wavelength λ of the neutron is

$$\lambda = \frac{h}{p}$$

$$= \frac{h}{\sqrt{2mE}}$$

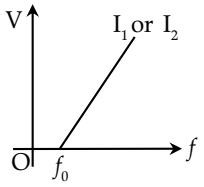
$$= \frac{6.63\times10^{-34}}{\sqrt{2\times1.67\times10^{-27}\times1.6\times10^{-19}}}$$

$$= 2.87 \times 10^{-11}\ \text{m}$$

$$\approx 10^{-8}\ \text{cm}$$

19. Option (4) is correct.

The value of stopping potential depends on the maximum kinetic energy of the photo-electron and thus it is independent of the intensity of incident light. Therefore, none of the graph shown is correct. The correct graph is shown in the figure.



20. Option (4) is correct.

Phosphorus is pentavalent impurity. Its doping will not affect the concentration of holes. So, the number of holes will be the same as in the intrinsic semiconductor. So, $n_h = 1.41 \times 10^{16}\ \text{m}^{-3}$.

21. Correct answer is [30°].

$$i = \frac{3}{4}A$$

$\Rightarrow$ $$i = 45°$$

and $$\delta_m = 2i - A$$

$$= 2 \times 45° - 60°$$

$\Rightarrow$ $$i = 30°$$

22. Correct answer is [375].

In an adiabatic process,

$$TV^{\gamma-1} = \text{constant}$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\frac{300}{T_2} = \left(\frac{\frac{8}{27}V}{V}\right)^{\frac{5}{3}-1}$$

$$= \left(\frac{8}{27}\right)^{\frac{2}{3}}$$

$$= \frac{4}{9}$$

$$T_2 = 300 \times \frac{9}{4}$$

$$= 675\ \text{K}$$

$\therefore$ Rise in temperature

$$= 675 - 300$$

$$= 375\ \text{K}$$

23. Correct answer is [5.00].

If C doesn't move then

$$a_A = 4a_B \quad \ldots(1)$$

$$P - T - \mu mg = m\,4a_B \quad \ldots(2)$$

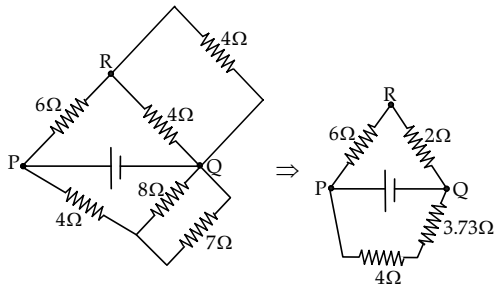
$$4T - mg = ma_B \quad \ldots(3)$$

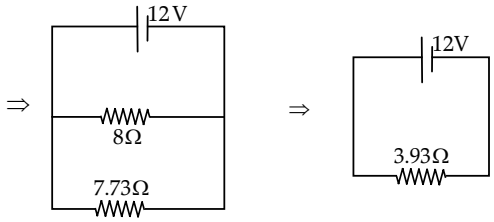
$$2T = mg \quad \ldots(4)$$

Solving above equations

$\therefore$ $$\frac{P}{mg} = 5$$

24. Correct answer is [36.00].

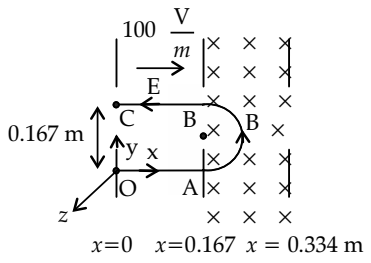




$$\text{Power dissipated} = \frac{(12)^2}{3.93} = 36 \text{ W}$$

25. Correct answer is [7.07].

The situation described in the problem is shown in the figure. As electric field is along x-axis, proton will be accelerated by the electric field and will enter the magnetic field at A (i.e., $x = 0.167$, $y = 0$) with velocity v along x-axis such that



$$\frac{1}{2}mv^2 = W = Fd = qEd$$

i.e., $$v = \left[\frac{2qEd}{m}\right]^{1/2}$$

$$= \left[\frac{2\times1.6\times10^{-19}\times100\times0.167}{1.67\times10^{-27}}\right]^{1/2}$$

$$= 4\sqrt{2}\times10^4 \frac{\text{m}}{\text{s}}$$

Now, as the proton is moving perpendicular to the magnetic field, it will describe a circular path in the magnetic field with radius r, such that

$$r = \frac{mv}{q\text{B}}$$

And as it comes back at C [$x = 0$; $y = 0.167$ m], its path in the magnetic field will be a semicircle such that

$$y = 2r = \frac{2mv}{q\text{B}}$$

i.e., $$\text{B} = \frac{2mv}{qy}$$

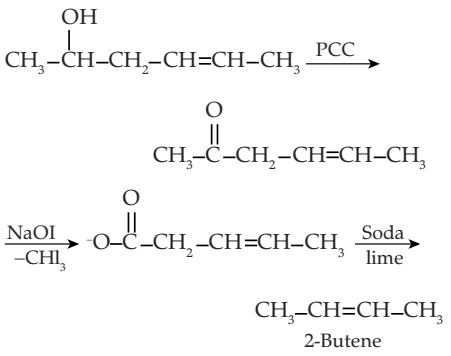
i.e., $$\text{B} = \frac{2\times1.67\times10^{-27}\times4\sqrt{2}\times10^4}{1.6\times10^{-19}\times0.167}$$

$$= \frac{1}{\sqrt{2}}\times10^{-2}$$

$$= 7.07 \text{ mT}$$

Chemistry

26. Option (3) is correct.



27. Option (1) is correct.

$$N_2(14) = (\sigma_{1s})^2(\sigma^*_{1s})^2(\sigma_{2s})^2(\sigma^*_{2s})^2$$

$$\left(\pi_{2p_x^2} = \pi_{2p_y^2}\right)(\sigma_{2p_z})^2$$

Bond order = $\frac{1}{2}$ (No. of electrons in bonding orbital —No. of electrons in anti bonding orbital)

$$\text{B.O} = \frac{1}{2}(10-4) = 3$$

$$NO^+(14) = (\sigma_{1s})^2(\sigma^*_{1s})^2(\sigma_{2s})^2(\sigma^*_{2s})^2$$

$$\left(\pi_{2p_x^2} = \pi_{2p_y^2}\right)(\sigma_{2p_z})^2$$

$$\text{B.O} = \frac{1}{2}(10-4) = 3$$

28. Option (1) is correct.

Given,

K_{sp} for bismuth sulphide (Bi_2S_3)

$$= 1.08\times10^{-7}$$

$$K_{sp}(Bi_2S_3) = [Bi^{3+}]^2\,[S^{2-}]^3$$

$$= (2S)^2\times(3S)^3$$

$$K_{sp} = 108\,S^5$$

$$S^5 = 1.08\times10^{-73}/108$$

$$S = 10^{-15}$$

29. Option (1) is correct.

The electronic configuration of these elements are:

Z = 107 [Rn] $5f^{14}\,6d^5\,7s^2$

Z = 108 [Rn] $5f^{14}\,6d^6\,7s^2$

Z = 109 [Rn] $5f^{14}\,6d^7\,7s^2$

These elements will be place in *d*-block in groups 7th, 8th and 9th respectively as they have incompletely filled *d*-orbital.

30. Option (4) is correct.

$H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2} - Ka_3$

$H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4 - Ka_1$

$HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2} - Ka_2$

$$K_{a_3} = \frac{\left[H^+\right]^2\left[C_2O_4^{2-}\right]}{\left[H_2C_2O_4\right]}$$

$$K_{a_1} = \frac{\left[H^+\right]\left[HC_2O_4^-\right]}{\left[H_2C_2O_4\right]}, K_{a_2} = \frac{\left[H^+\right]\left[C_2O_4^{2-}\right]}{\left[HC_2O_4^-\right]}$$

$K_{a_3} = K_{a_1} \times K_{a_2}$

31. Option (2) is correct.

$\Delta H_{fusion} = 1.435 \text{ kcal mol}^{-1}$

$= 1435 \text{ cal mol}^{-1}$

$\Delta S = 5.26 \text{ cal mol}^{-1} \text{ K}^{-1}$

$$\Delta S = \frac{\Delta H_{fusion}}{T}$$

$$T = \frac{\Delta H_{fusion}}{\Delta S} = \frac{1435}{5.26}K$$

$= 272.81K \approx 0°C$

32. Option (2) is correct.

$Fe^{+3}\,[Fe\,(CN)_6]^{4-}$

Let the oxidation state of Fe in the complex be x

$x + 6\,(-1) = -4$

$x = -4 + 6$

$x = +2$

33. Option (2) is correct.

Electronic configuration of O_2^- (17 electrons) is as follows:

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^{\,2} < \pi 2p_x^{\,2}$

$= \pi 2p_y^{\,2} < \pi^* 2p_x^{\,2} = \pi^* 2p_y^{\,1}$

It contains 1 unpaired electron thus, it is paramagnetic in nature.

34. Option (4) is correct.

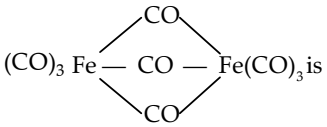
The ionisation energy of Cu is more than that of K though both have a 4*s* configuration. This is because Cu has filled the 3*d* orbital which increases the attraction between the nucleus and electrons.

35. Option (2) is correct.

$CH_3CH_2–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{N}}–CH=CH_2$

Ethyl methyl vinyl amine

36. Option (1) is correct.

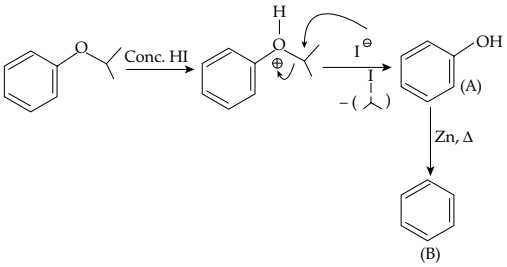


Tri-μ-carbonyl bis (tricarbonyl ion (0))

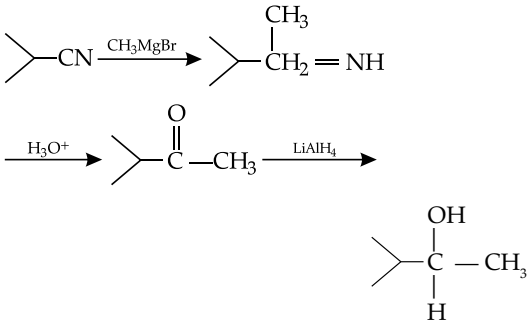
37. Option (4) is correct.

Coordination compounds having coordination number 6 of type MA_4B_2, MA_3B_3, $M(AB)_3$, $M(AA)_2B_2$ exhibit geometrical isomerism.

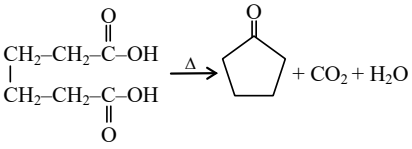
38. Option (4) is correct.



39. Option (3) is correct.



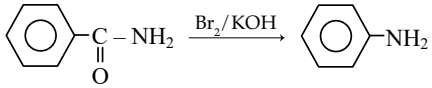
40. Option (2) is correct.



Adipic acid Cyclopentanone

Adipic acid undergo dehydration as well as decarboxylation to give cyclopentanone when it is heated.

41. Option (2) is correct.



Benzanide (A) Aniline

42. Option (2) is correct.

$(C_6H_{12}O_5)n + nH_2O \xrightarrow[\text{diastase}]{(x)} C_{12}H_{22}O_{11}$

$C_{12}H_{22}O_{11} + H_2O \xrightarrow[H_2O]{\text{Maltase (y)}} 2C_6H_{12}O_6$

43. Option (4) is correct.

From photoelectric Einstein equation:

$$E = E^\circ + K.E.$$

$$h\nu = h\nu^\circ + K.E.$$

$$K.E. = h\nu - h\nu^\circ$$

$$K.E. = h(\nu - \nu^\circ)$$

If $\nu > \nu_0$, K.E. > 0

So, if frequency of the light (ν) increases the K.E. of these electrons increases.

44. Option (1) is correct.

In the Hunsdiecker reaction →

$$CH_3CH_2COOAg + Br_2 \xrightarrow{CCl_4}$$

Silver Propionate (No. of C = 3)

$$CH_3CH_2Br + CO_2 + AgBr$$

Ethyl Bromide (No. of C = 2)

No. of carbon atoms decrease.

45. Option (3) is correct.

$$CaC_2 + 2H\text{–}OH \rightarrow Ca(OH)_2 + H - C \equiv C - H$$

(A) Acetylene

$\downarrow Hg^{2+}/H_2SO_4$

$$CH_3\text{–}CH_2OH \xleftarrow{H_2/Ni} H - \underset{\underset{O}{\|}}{C} - CH_3$$

(C) Ethanol — (B) Acetaldehyde

46. Correct answer is [10.70].

$$\mu = \frac{K_{T+10}}{K} = 1.75$$

$$\log \frac{K_{T+10}}{K} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$T_1 = 25°C = 298K$

$T_2 = 35°C = 308K$

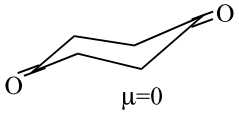
$$2.303 \times \log \frac{K_{T+10}}{K} = \frac{E_a}{1.987}\left[\frac{308 - 298}{308 \times 298}\right]$$

$$E_a = \frac{2.303 \times 308 \times 298 \times 1.987}{10} \log 1.75 \text{ cal. mol}^{-1}$$

$$= 10.2 \text{ kcal mol}^{-1}$$

$$E_a = 10.2 \text{ kcal mol}^{-1}$$

47. Correct answer is [6.0].



μ=0

Twist boat μ = 0

$$\mu_{net} = \mu_{chair}\, x_{chair} + \mu_{t.b.}\, x_{t.b.}$$

$$1.2 = 0 + \mu_{t.b.}\,(0.20)$$

$\because$ $x_{chair} = 0.80$

$\therefore$ $x_{t.b.} = 1 - 0.80 \Rightarrow 0.2$

$\Rightarrow$ $\mu_{t.b.} = 6.0$ debye

48. Correct answer is [266].

Given, $R = 1750\Omega$

$K = 0.152 \times 10^{-3}\ \Omega^{-1}cm^{-1}$

Conductivity = Cell constant/ Resistance

Cell constant = Conductivity × Resistance

$= 0.152 \times 10^{-3}\ \Omega^{-1}cm^{-1} \times 1750\Omega$

Cell constant $= 266 \times 10^{-3}\ cm^{-1}$

The cell constant depends on the distance between the electrodes and their area of cross–section

49. Correct answer is [112.70].

$$\frac{P^o - P_s}{P_s} = \frac{n}{N}, \rightarrow x_2 \text{ (mole fraction)}$$

$$\frac{P^o - P_s}{P_s} = \frac{n_2}{n_1} (\text{since } n_1 >> n_2)$$

$$= \frac{W_2}{M_2} \times \frac{M_2}{W_1}$$

$$\frac{736 - 526}{526} = \frac{2.5x}{M_2} \times \frac{18}{1x}$$

$$M_2 = 112.7 \text{ g}$$

50. Correct answer is [1.122].

$$Ph - MgBr + HC \equiv C - CH_2\text{–}CH_2\text{–}CH_2\text{–}OH$$

$\downarrow$

$$C_6H_6 + HC \equiv C\text{–}CH_2CH_2CH_2OMgBr$$

since, 84 g $HC \equiv C\text{–}CH_2\text{–}CH_2CH_2\text{–}OH$

gives → 22.4 L Benzene

$\therefore$ 1 g of Pent – 4 – yne – 1 – ol → $\frac{22.4}{84}$

$\therefore$ 4.2 g of Pent – 4 – yne – 1 – ol →

$$\frac{22.4}{84} \times 4.2 = 1.12$$

Mathematics

51. Option (2) is correct.

$$32^{32} = (2^5)^{32} = 2^{160}$$

$$= (3 - 1)^{160} = 3k + 1$$

Now, $32^{32^{32}} = (2^5)^{3k+1} = 2^{(15k+5)}$

$$= 2^{(15k+3)+2} = 4.2^{3(5k+1)}$$

$$= 4.8^{5k+1} = 4(7 + 1)^{\alpha}$$

$$= 4[7\mu + 1]$$

$\therefore$ The remainder when $32^{32^{32}}$ is divided by 7 is 4.

So, $m = 4$

Now, $\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}} = e^n$

$\Rightarrow \quad e^n = e^{\lim\limits_{x\to 0}\left[1+\frac{\sin x - x}{x}\right]\frac{\sin x}{x-\sin x}}$

$\Rightarrow \quad n = \lim\limits_{x\to 0}\left(-\frac{\sin x}{x}\right)$

$\Rightarrow \quad n = -1$

Now, the quadratic equation whose roots are $\frac{1}{m}$ and $\frac{1}{n}$ is given by $x^2 - \left(\frac{1}{m}+\frac{1}{n}\right)x + \frac{1}{mn} = 0$.

$\Rightarrow \quad x^2 - \left(\frac{1}{4}-1\right)x - \frac{1}{4} = 0$

$\Rightarrow \quad x^2 + \frac{3}{4}x - \frac{1}{4} = 0$

$\Rightarrow \quad 4x^2 + 3x - 1 = 0$

Hint:

(1) First, simplify 32^{32} using binomial theorem, then simplify $32^{32^{32}}$ and analyse further.

(2) Quadratic equation whose roots are α and β is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

52. Option (1) is correct.

Let $I = \int_0^1 \frac{dx}{((x-1)^2+1)^3}$

$= -\int_0^1 \frac{dx}{(x^2+1)^3}$

Let $x = \tan\theta$

$\Rightarrow \quad dx = \sec^2\theta\, d\theta$

$\int_0^{\pi/4} \frac{\sec^2\theta}{\sec^6\theta}\, d\theta$

$= \int_0^{\pi/4} \cos^4\theta\, d\theta$

$= \frac{1}{4}\int_0^{\pi/4} (1+\cos 2\theta)^2\, d\theta$

$= \frac{1}{4}\int_0^{\pi/4} (1+2\cos 2\theta+\cos^2 2\theta)\, d\theta$

$= \frac{1}{4}\int_0^{\pi/4} (1+2\cos 2\theta)\, d\theta + \frac{1}{8}\int_0^{\pi/4} (1+\cos 4\theta)\, d\theta$

$= \frac{1}{4}\left[\theta+\sin 2\theta\right]_0^{\frac{\pi}{4}} + \frac{1}{8}\left(\theta+\frac{\sin 4\theta}{4}\right)_0^{\frac{\pi}{4}}$

$= \frac{1}{4}\left[\frac{\pi}{4}+1\right] + \frac{1}{8}\left[\frac{\pi}{4}\right]$

$= \frac{3\pi+8}{32}$

53. Option (3) is correct.

Given system of linear equations:

$x + y + z = 6$

$x + 2y + 3z = 10$

$x + 2y + k_1 z = k_2$

Now, $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & k_1 \end{vmatrix}$

$= \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & k_1-2 \end{vmatrix} = k_1 - 3$

$\Delta_x = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 3 \\ k_2 & 2 & k_1 \end{vmatrix}$

$= \begin{vmatrix} 0 & 1 & 0 \\ -2 & 2 & 1 \\ k_2-12 & 2 & k_1-2 \end{vmatrix}$

$= 2k_1 + k_2 - 16$

$\Delta_y = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 10 & 3 \\ 1 & k_2 & k_1 \end{vmatrix}$

$= \begin{vmatrix} 1 & 6 & 1 \\ 0 & 4 & 2 \\ 0 & k_2-10 & k_1-3 \end{vmatrix}$

$= 4k_1 - 2k_2 + 8$

$\Delta_z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 10 \\ 1 & 2 & k_2 \end{vmatrix}$

$= \begin{vmatrix} 1 & 1 & 6 \\ 0 & 1 & 4 \\ 0 & 0 & k_2-10 \end{vmatrix}$

$= k_2 - 10$

For unique solution, $\Delta \neq 0 \Rightarrow k_1 \neq 3$

For infinite number of solutions,

$\Delta = 0, \Delta_x = \Delta_y = \Delta_z = 0$

$\therefore \quad k_1 = 3$ and $k_2 = 10$

For no solution, $\Delta = 0$ and at least one of Δ_x, Δ_y, Δ_z is non zero.

$\therefore \quad k_1 = 3$ and $k_2 = 10$

Hint:

(1) Find Δ, Δ_x, Δ_y and Δ_z in terms of k_1 and k_2.

(2) Unique solution, $\Delta \neq 0$.

(3) For no solution, $\Delta = 0$ and at least one of Δ_x, Δ_y, Δ_z is non zero and for infinite solutions,

$$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$$

54. Option (1) is correct.

Given differential equation, $\frac{dy}{dx}(x^2y^3 + xy) = 1$

$$\Rightarrow \quad \frac{dx}{dy} = x^2y^3 + xy$$

$$\Rightarrow \quad \frac{dx}{dy} - xy = x^2y^3$$

$$\Rightarrow \quad \frac{1}{x^2}\frac{dx}{dy} - \frac{y}{x} = y^3$$

Let $-\frac{1}{x} = u$

$$\Rightarrow \quad \frac{1}{x^2}\frac{dx}{dy} = \frac{du}{dy}$$

So, $\frac{du}{dy} + uy = y^3$, which is a linear differential equation.

Now, I.F. $= e^{\int y dy} = e^{y^2/2}$

So, solution of given differential equation is

$$ue^{y^2/2} = \int y^3 e^{y^2/2} dy + c_1$$

$$\Rightarrow \quad ue^t = \int 2te^t dt + c_1 \quad \left\{\text{Putting } \frac{y^2}{2} = t\right\}$$

$$\Rightarrow \quad ue^t = 2e^t(t-1) + c_1$$

$$\Rightarrow \quad u = 2(t-1) + c_1e^{-t}$$

$$\Rightarrow \quad -\frac{1}{x} = 2\left(\frac{y^2}{2} - 1\right) + c_1e^{-y^2/2}$$

$$\Rightarrow \quad -1 = x(y^2 - 2) + xe^{-y^2/2}c_1$$

$$\Rightarrow \quad x(y^2 - 2 + c_1e^{-y^2/2}) = -1$$

$$\therefore \quad m = 2$$
$$n = -2$$
$$k = 2$$
$$l = -1$$

So, ${}^{m-n+k+l}C_m = {}^5C_2 = \frac{5.4}{2.1} = 10$

Hint: Simplify the given differential equation, substitute $-\frac{1}{x} = u$ and solve further using the concept of linear differential equation.

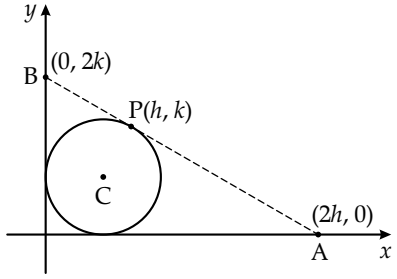
55. Option (1) is correct.

Given equation of circle is,

$$x^2 + y^2 - 2\lambda x - 2\lambda y + \lambda^2 = 0$$

$\Rightarrow$ Centre $= (\lambda, \lambda)$

and Radius $= \lambda$



Let the circumcentre be P(h, k)

As we know, circumcentre of a right angle triangle is the midpoint of hypotenuse.

$\therefore$ A $= (2h, 0)$

and B $= (0, 2k)$

Now, equation of AB is $\frac{x}{2h} + \frac{y}{2k} = 1$

$\because$ AB is tangent to circle

$\Rightarrow$ CP $= \lambda$

$$\Rightarrow \quad \lambda = \frac{\left|\frac{\lambda}{2h} + \frac{\lambda}{2k} - 1\right|}{\sqrt{\frac{1}{4h^2} + \frac{1}{4k^2}}}$$

$$\Rightarrow \quad \frac{\lambda}{2hk}\sqrt{h^2 + k^2} = |\lambda k + \lambda h - 2hk|\frac{1}{2hk}$$

$$\Rightarrow \quad \lambda^2(h^2 + k^2) = \lambda^2k^2 + \lambda^2h^2 + 4h^2k^2 + 2\lambda^2kh - 4\lambda h^2k - 4hk^2\lambda$$

$$\Rightarrow \quad 4\lambda hk(h + k) = 2hk(2hk + \lambda^2)$$

$$\Rightarrow \quad 2\lambda(h + k) = 2hk + \lambda^2$$

$$\Rightarrow \quad 2(h + k) = \frac{2hk}{\lambda} + \lambda$$

$$\Rightarrow \quad 2(x + y) = \frac{2xy}{\lambda} + \lambda$$

$$\Rightarrow \quad 2(x + y) - \lambda = \frac{2xy}{\lambda}$$

$$\Rightarrow \quad (m + n) = 4$$

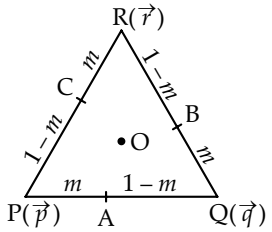
Hint:

(1) Circumcentre of right angle triangle is the mid point of hypotenuse.

(2) The perpendicular distance from the centre to the tangent is equal to the circle's radius.

56. Option (3) is correct.

Let $\vec{p}, \vec{q}, \vec{r}$ be the position vectors of P, Q, R respectively.



Given, $\frac{PA}{PQ} = \frac{QB}{QR} = \frac{RC}{RP} = m$

$\Rightarrow \quad \vec{OA} = m\vec{q} + (1-m)\vec{p}$

$\Rightarrow \quad \vec{OB} = m\vec{r} + (1-m)\vec{q}$

$\Rightarrow \quad \vec{OC} = m\vec{p} + (1-m)\vec{r}$

$\vec{PB} = \vec{OB} - \vec{OP}$

$= m\vec{r} + (1-m)\vec{q} - \vec{p}$

$\vec{QC} = m\vec{p} + (1-m)\vec{r} - \vec{q}$

$\vec{RA} = m\vec{q} + (1-m)\vec{p} - \vec{r}$

Now, area of triangle formed by $\vec{PB}$, $\vec{QC}$, $\vec{RA}$ is

$$\Delta = \frac{1}{2}|\vec{PB} \times \vec{QC} + \vec{QC} \times \vec{RA} + \vec{RA} \times \vec{PB}|$$

$$\Delta = \frac{1}{2}|m^2 - m + 1| \ (\Delta PQR)$$

So, Δ is least if $m = \frac{1}{2}$.

Hint: Assume position vector of P, Q, R, find $\vec{OA}$, $\vec{OB}$, $\vec{OC}$ and then find $\vec{PB}$, $\vec{QC}$ and $\vec{RA}$ and solve further.

57. Option (1) is correct.

$$[3 \quad x+2 \quad x+1]\begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$$

$\Rightarrow \quad 3x - (x+2) + (x+1) = 0$

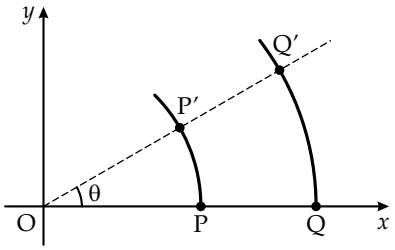
$\Rightarrow \quad 3x - 1 = 0$

$\Rightarrow \quad x = \frac{1}{3}$

58. Option (2) is correct.

Given, Points P $= (\sqrt{13}, 0)$

and $Q = \left(2\sqrt{13}, 0\right)$



$\because \quad \theta = \tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow \quad \tan\theta = \frac{2}{3}$

$\Rightarrow \quad \sin\theta = \frac{2}{\sqrt{13}}$

and $\cos\theta = \frac{3}{\sqrt{13}}$

$\Rightarrow \quad P' = (OP\cos\theta, OP\sin\theta)$

$= (3, 2)$

Similarly, $Q' = (OQ\cos\theta, OQ\sin\theta)$

$= (6, 4)$

Now, equation of P′Q′ is

$\Rightarrow \quad y - 2 = \left(\frac{4-2}{6-3}\right)(x-3)$

$\Rightarrow \quad y - 2 = \left(\frac{2}{3}\right)(x-3)$

$\Rightarrow \quad 3y - 6 = 2x - 6$

$\Rightarrow \quad 3y - 2x = 0$

Hence, foot of perpendicular from (4, –2) to line P′Q′ is (α, β)

$\because \quad \frac{\alpha-4}{-2} = \frac{\beta+2}{3}$

$= \frac{-(3(-2)-2(4))}{(3)^2+(2)^2}$

$\frac{\alpha-4}{-2} = \frac{\beta+2}{3} = +\frac{14}{13}$

$\Rightarrow \quad \alpha = \frac{-28}{13} + 4 = \frac{24}{13}$

and $\beta = \frac{42}{13} - 2 = \frac{16}{13}$

$\Rightarrow \quad 13(\alpha + \beta) = 24 + 16 = 40$

Hint:

(1) Use P′ = (OP cos θ, OP sin θ);

where $\theta = \tan^{-1}\left(\frac{2}{3}\right)$

(2) If P (x_2, y_2) is the foot of a perpendicular drawn from Q (x_1, y_1) on the line

$ax + by + c = 0$

then $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = -\frac{(ax_1 + by_1 + c)}{a^2 + b^2}$

59. Option (2) is correct.

Total cases = 52

Favourable cases $x = 13 + 3 = 16 = x$

Unfavourable cases $y = 52 - 16 = 36 = y$

Odds in favour $= \frac{x}{y} = \frac{16}{36} = \frac{4}{9} = 4:9$

60. Option (3) is correct.

Let $z = x + iy$

Given, $\arg\left(\frac{z - z_1}{z - z_2}\right) = \frac{\pi}{2}$

$\Rightarrow$ z lines on the circle with z_1, z_2 as extrimities of a diameter.

$\Rightarrow$ Centre (7, 6) and radius = 3

$\therefore$ $|z - 7 - 9i| = 3\sqrt{2}$

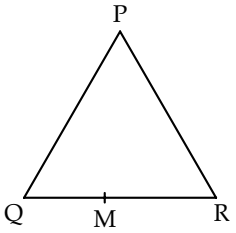
61. Option (4) is correct.

Given, $P = (2, 3, 5)$

$Q = (-1, 2, 2)$

and $R = (\alpha, 5, \beta)$

mid point of $QR = \left(\frac{\alpha - 1}{2}, \frac{7}{2}, \frac{\beta + 2}{2}\right)$



Direction ratios of PM

$= \frac{\alpha - 1}{2} - 2, \frac{7}{2} - 3, \frac{\beta + 2}{2} - 5$

$\because$ PM is equally inclined to the axis.

$\frac{\alpha - 5}{2} = \frac{1}{2} = \frac{\beta - 8}{2}$

$\Rightarrow$ $\alpha = 6$

and $\beta = 9$

So, $n = \alpha + \beta = 15$

Let $a = {}^nC_0 + 3{}^nC_1 + 5{}^nC_2 + \ldots + (2n + 1)\,C_n$

$a = \sum_{r=0}^{n}(2r+1)C_r$

$= 2\sum_{r=0}^{n} r\,C_r + \sum_{r=0}^{n} C_r$

$= 2(n\,2^{n-1}) + 2^n$

$= 2^n (n + 1)$

$\Rightarrow$ $a = 2^{15}(15 + 1) = 2^{19}$

Hint:

(1) If line $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$ is equally inclined to axes, then $a = b = c$.

(2) $\sum_{r=0}^{n} r\,C_r = n2^{n-1}$ and $\sum_{r=0}^{n} C_r = 2^n$

62. Option (2) is correct.

Given, $g : R \to \left(0, \frac{\pi}{2}\right]$; $g(x) = \cot^{-1}(x^2 + x + k)$

$\because$ $g(x)$ is an onto function.

$\Rightarrow$ $0 < \cot^{-1}(x^2 + x + k) \le \frac{\pi}{2}$

$\Rightarrow$ $0 \le x^2 + x + k < \infty$

$\Rightarrow \left(x + \frac{1}{2}\right)^2 + \left(k - \frac{1}{4}\right) \ge 0$

This is possible $\forall\ x \in R$, iff $k - \frac{1}{4} = 0$

$\Rightarrow$ $k = \frac{1}{4}$

Hint: For an onto function, codomain = range.

63. Option (2) is correct.

- a, b, c are in AP

$\Rightarrow$ $2b = a + c$

$\Rightarrow$ $2ab = a^2 + ac$

$\Rightarrow$ $a^2 - 2ab = -ac$

- a, b, d in GP

$\Rightarrow$ $b^2 = ad$

$\Rightarrow$ $a^2 - 2ab + b^2 = -ac + ad$

$\Rightarrow$ $(a - b)^2 = a(d - c)$

$\Rightarrow$ $a, a - b, d - c$ are in GP

64. Option (3) is correct.

$$\frac{1}{1!\cdot(n-1)!} + \frac{1}{3!\cdot(n-3)!} + \frac{1}{5!\cdot(n-5)!} + \ldots$$

$$= \frac{1}{n!}\left[\frac{n!}{1!(n-1)!} + \frac{n!}{3!(n-3)!} + \frac{n!}{5!(n-5)!} + \ldots\right]$$

$$= \frac{1}{n!}\left[{}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\right]$$

Since,

${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + {}^nC_4 + {}^nC_5 + \ldots + {}^nC_n = 2^n$

And ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 - {}^nC_5 + \ldots + (-1)^n\,{}^nC_n = 0$

Subtracting, we get

$2[{}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots] = 2^n$

$\Rightarrow$ ${}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots = \frac{2^n}{2} = 2^{n-1}$

$\therefore$ $\frac{n!}{1!(n-1)!} + \frac{n!}{3!(n-3)!} + \frac{n!}{5!(n-5)!} + \ldots = 2^{n-1}$

$\Rightarrow$ $\left[\frac{1}{1!(n-1)!} + \frac{1}{3!(n-3)!} + \frac{1}{5!(n-5)!} + \ldots\right] = \frac{2^{n-1}}{n!}$

65. Option (4) is correct.

Let $I = \int \frac{\cos^2 x + \sin 2x}{(2\cos x - \sin x)^2} dx$

$\Rightarrow \quad I = \int \frac{1 + 2\tan x}{(2 - \tan x)^2}$

$\Rightarrow \quad I = \int \frac{(\sec^2 x - \tan^2 x) + 2\tan x}{(2 - \tan x)^2} dx$

$\Rightarrow \quad I = \int \frac{\sec^2 x}{(2 - \tan x)^2} dx + \int \frac{(2 - \tan x)\tan x}{(2 - \tan x)^2} dx$

$\Rightarrow \quad I = \int \frac{\sec^2 x}{(2 - \tan x)^2} dx + \int \frac{\tan x\ (1 + \tan^2 x)}{(2 - \tan x)(1 + \tan^2 x)} dx$

Let $\tan x = u \Rightarrow \sec^2 x\, dx = du$

$\Rightarrow \quad I = \int \frac{du}{(2-u)^2} + \int \frac{u\,du}{(2-u)(1+u^2)}$

$\Rightarrow \quad I = \frac{1}{2-u} - \frac{2}{5}\log_e(2-u) + \frac{1}{5}\log_e(1+u^2)$

$$-\frac{1}{5}\tan^{-1} u + c$$

$\Rightarrow \quad I = \frac{1}{2 - \tan x} - \frac{2}{5}\log_e(2 - \tan x)$

$$+\frac{1}{5}\log_e \sec^2 x - \frac{1}{5}x + c$$

$\therefore \quad f(x) = 2 - \tan x$

$g(x) = \sec^2 x$

and $h(x) = x$

$m = 2$

$n = 1$

$r = 1$

So $f'(0) + g'(0) + h'(0) = 0$

Hint: Divide the numerator and denominator by $\cos^2 x$, simplify using $1 = \sec^2 x - \tan^2 x$ and substitute $\tan x = u$ and solve further.

66. Option (2) is correct.

Given, $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

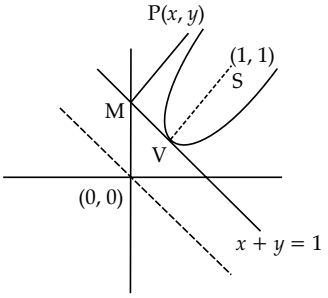
$B = \{1, 3, 4, 5, 6\}$

$\Rightarrow \quad B^C = \{2, 7, 8, 9, 10, 11\}$

$\therefore$ Number of subset of B^C which contain 10 is 2^5.

Hint: The total number of subset of finite set containing n elements is 2^n.

67. Option (1) is correct.



Distance between focus and vertex = a

$\Rightarrow \quad a = \frac{1+1-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

Now, equation of directrix is $x + y = 0$

Since, $SP^2 = PM^2$

$\Rightarrow \quad (x-1)^2 + (y-1)^2 = \left(\frac{x+y}{\sqrt{2}}\right)^2$

$\Rightarrow \quad x^2 + y^2 - 2x - 2y + 1 + 1 = \frac{(x+y)^2}{2}$

$\Rightarrow \quad 2(x^2 + y^2 - 2x - 2y + 2) = x^2 + y^2 + 2xy$

$\Rightarrow \quad x^2 + y^2 - 2xy - 4x - 4y + 4 = 0$

68. Option (3) is correct.

Given, $f(x) = \frac{1}{\sqrt{x - 2\{x\}}} + \ln\{x\}$

Domain of $\ln\{x\}$ is R – I

$\frac{1}{\sqrt{x - 2\{x\}}}$ is exist if $x - 2\{x\} > 0$

$\Rightarrow \quad x > 2\{x\}$

$\Rightarrow \quad x \in [1, \infty]$

$\therefore$ Domain of $f(x)$ is $[1, \infty) - I^+$

Hint:

(1) $\ln x$ is defined for $x > 0$

(2) $\frac{1}{\sqrt{g(x)}}$ is defined for $g(x) > 0$

69. Option (3) is correct.

Given, hyperbolas,

H: $x^2 + 3xy + 2y^2 + 2x + 3y + 2 = 0$

and H^1: $x^2 + 3xy + 2y^2 + 2x + 3y + c = 0$

Since, H and H^1 are conjugate of each other.

$\therefore$ $H + H^1 = 0$, represents pair of straight lines.

So, $2(x^2 + 3xy + 2y^2 + 2x + 3y) + c + 2 = 0$ represents pair of straight line.

As we know, if $ax^2 + 2hxy + by^2 + 2gx + 2fy + c_1 = 0$ represents pair of straight lines, then

$abc_1 + 2fgh - af^2 - bg^2 - c_1h^2 = 0$

$\therefore$ (2) (4) (c + 2) +2 (3) (2) (3) – 2 $(3)^2$ – 4 $(2)^2$ – (c + 2) $(3)^2$ = 0

$\Rightarrow c = 0$

70. Option (3) is correct.

Total ways of selecting 5 numbers from 1, 2, 3.....18 is

$$n(S) = {}^{18}C_5$$

Arrange in the increasing order

$$x_1 < x_2 < x_3 < x_4 < x_5$$
$$\downarrow \quad \downarrow$$
$$7 \quad 11$$

choice for x_1 are {1, 2, 3, 4, 5, 6}

choice for x_3 are {8, 9, 10}

choice for x_5 are {12, 13, 14, 15, 16, 17, 18}

$$\therefore \quad n(E) = {}^6C_1 \times {}^3C_1 \times {}^7C_1$$
$$= 6 \times 3 \times 7$$

$$\therefore \quad P(E) = \frac{6\times3\times7}{{}^{18}C_5} = \frac{1}{68}$$

71. Correct answer is [80].

Given

$$\Rightarrow \sin^2 10^\circ.\sin 20^\circ.\sin 40^\circ \sin 50^\circ \sin 70^\circ = \alpha - \frac{1}{16}\sin 10^\circ$$

We have by trigonometric identity

$$\Rightarrow \sin\alpha.\sin(60^\circ - \alpha)\sin(60 + \alpha) = \frac{1}{4}\sin 30^\circ$$

$$\Rightarrow \sin 10^\circ.\frac{1}{2}(2\sin 20^\circ \sin 40^\circ)\sin 10^\circ \sin 50^\circ \sin 70^\circ = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{2}\sin 10^\circ(\cos(40^\circ - 20^\circ) - \cos(40^\circ + 20^\circ))$$
$$\sin 10^\circ \sin(60^\circ - 10^\circ)\sin(60^\circ + 10^\circ) = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{2}\sin 10^\circ(\cos 20^\circ - \cos 60^\circ).\frac{1}{4}\sin 30^\circ = \alpha - \frac{1}{6}\sin 10^\circ$$

$$\Rightarrow \frac{1}{8}\sin 10^\circ\left(\cos 20^\circ - \frac{1}{2}\right)\times\frac{1}{2} = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{16}\sin 10^\circ\left(\cos 20^\circ - \frac{1}{2}\right) = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{32}\sin 10^\circ(2\cos 20^\circ - 1) = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{32}[2\sin 10^\circ \cos 20^\circ - \sin 10^\circ] = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{32}[\sin(30^\circ) + \sin(10^\circ - 20^\circ) - \sin 10^\circ] = \alpha - \frac{1}{16}\sin 10^\circ$$

$\{\because 2\sin A\sin B = \cos(A - B) - \cos(A + B), 2\sin A\cos B = \sin(A + B) + \sin(A - B)\}$

$$\Rightarrow \frac{1}{32}\left[\frac{1}{2} - \sin 10^\circ - \sin 10^\circ\right] = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{64} - \frac{2}{32}\sin 10^\circ = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\Rightarrow \frac{1}{64} - \frac{1}{16}\sin 10^\circ = \alpha - \frac{1}{16}\sin 10^\circ$$

$$\therefore \quad \alpha = \frac{1}{64}$$

Now, the value of $16 + \alpha^{-1} = 16 + \frac{1}{\alpha}$

$= 16 + 64 = 80$

72. Correct answer is [8.00].

$$\frac{f(\pi/4)}{1/2} = -\cos^{-1}\left(\frac{1}{\sqrt{2}}\right) - \left|f\left(\frac{\pi}{4}\right)\right|$$

$$\Rightarrow 2f\left(\frac{\pi}{4}\right) = -\left(\frac{\pi}{4} + \left|f\left(\frac{\pi}{4}\right)\right|\right)$$

$\Rightarrow f\left(\frac{\pi}{4}\right)$ is a negative quantity

$$\Rightarrow 2f\left(\frac{\pi}{4}\right) = -\frac{\pi}{4} + f\left(\frac{\pi}{4}\right)$$

$$\Rightarrow f\left(\frac{\pi}{4}\right) = -\frac{\pi}{4}$$

$$\Rightarrow \frac{f\left(\frac{\pi}{4}\right)}{\left(-\frac{\pi}{32}\right)} = \frac{-\frac{\pi}{4}}{-\frac{\pi}{32}} = 8$$

73. Correct answer is [8.00].

$$\lim_{x\to 2}\frac{2^x + 2^{3-x} - 6}{\sqrt{2^{-x}} - 2^{1-x}} = A \text{ (let)}$$

$$\Rightarrow A = \lim_{x\to 2}\frac{2^x + 8.2^{-x} - 6}{2^{-x/2} - 2^{1-x}} \quad \left(\frac{0}{0} form\right)$$

So, by L' hospital rule, we get,

$$A = \lim_{x\to 2}\frac{2^x \log 2 + 8.2^{-x}\log 2(-1) - 0}{2^{-\frac{x}{2}}\left(-\frac{1}{2}\right)\log 2 - 2.2^{-x}\log 2(-1)}$$

$$\Rightarrow A = \lim_{x\to 2}\frac{2^x - 8.2^{-x}}{-2^{-\frac{x}{2}-1} + 2^{1-x}} = \frac{2^2 - 8/4}{-2^{-2} + 2^{-1}}$$

$$= \frac{4-2}{-\frac{1}{4}+\frac{1}{2}}$$

$$= \frac{2}{1/4} = 8$$

74. Correct answer is [11].

Given, $|z_1| = 6$

$\Rightarrow$ z_1 lies on circle $x^2 + y^2 = 6^2$

and $|z_2 - 4| + |z_2 + 4| = 10$

$\Rightarrow$ z_2 lies on ellipse whose length of major axis is 10 and having foci at $(\pm 4, 0)$

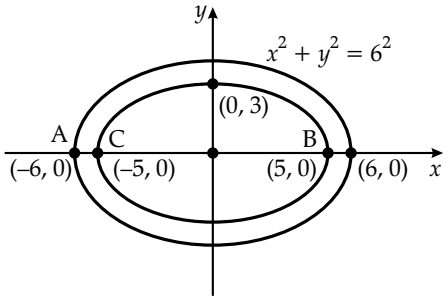
$\Rightarrow$ z_2 lies on ellipse $\frac{x^2}{5^2}+\frac{y^2}{3^2} = 1$

So, $|z_1 - z_2|_{\max} = AB = 11$

Hint:

(1) $|z - z_1| = r$, represent circle of radius r having centre at z_1.

(2) $|z - z_1| + |z - z_2| = 2a$, represents equation of ellipse whose length of major axis is $2a$ and having foci at z_1 and z_2.



75. Correct answer is [2.00].

$$(e-1)e^{xy} + x^2 = e^{x^2+y^2}$$

$$(e-1)e^{xy}\{xy' + y\} + 2x = e^{x^2+y^2}(2x + 2yy')$$

$\Rightarrow$ $$y' = \frac{(e-1)y.e^{xy} + 2x - 2x.e^{x^2+y^2}}{2ye^{x^2+y^2} - (e-1)e^{xy}.x}$$

$\Rightarrow$ $$\left.\frac{dy}{dx}\right|_{(1,0)} = \frac{0+2-2\cdot e}{0-(e-1)}$$

$$= \frac{2(1-e)}{1-e} = 2$$

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